

Q1. (Based on Exercise 3.1 from slides). Let $f(x) = \sin(\pi x/2)$ and let S be its PCHIP Spline interpolant at the nodes $\{x_i\}_{i=0}^3 = \{0, 1, 2, 3\}$

- (a) Evaluate S at the point $x = 2.5$.
- (b) Compute the (actual) the error at $x = 2.5$.
- (c) Compare this result with the error bound provided in Theorem 3.2

Answer:

- (a) Note: we are not asked to write down the formula for the spline, just to evaluate it at $x = 2.5$. The PCHIP interpolant can be written as

$$S(x) = \begin{cases} S_1(x) & 0 \leq x \leq 1 \\ S_2(x) & 1 \leq x \leq 2 \\ S_3(x) & 2 \leq x \leq 3 \end{cases}$$

So $S(2.5) = S_3(2.5)$. So we'll give the formula just for S_3 (and not S_1 or S_2). Let

$$S_3(x) = c_0 + c_1(x-2) + c_2(x-2)^2 + c_3(x-2)^3,$$

where

$$\begin{aligned} c_0 &= f_2, & c_1 &= f'_2, \\ c_2 &= \frac{3}{h^2}(f_3 - f_2) - \frac{1}{h}(f'_3 + 2f'_2), & c_3 &= \frac{1}{h^2}(f'_3 + f'_2) - \frac{2}{h^3}(f_3 - f_2). \end{aligned}$$

For this problem $h = 1$, $f_2 = f(2) = 0$, and $f_3 = f(3) = -1$. Also $f'(x) = \frac{\pi}{2} \cos(\pi x/2)$. So $f'_2 = f'(2) = -\pi/2$, and $f'_3 = f'(3) = 0$. With a bit of calculation,

$$c_0 = -1, \quad c_1 = -\pi/2,$$

$$c_2 = 3(f_3 - f_2) - (f'_3 + 2f'_2) = \pi - 3, \quad c_3 = (f'_3 + f'_2) - 2(f_3 - f_2) = 2 - \pi/2.$$

That gives

$$S_3(x) = -\frac{\pi}{2}(x-2) + (\pi-3)(x-2)^2 + (2-\pi/2)(x-2)^3.$$

Then we get that $S(2.5) = S_3(2.5) \approx -0.69635$.

- (b) The error at $x = 2.5$ is

$$|f(2.5) - S_3(2.5)| = |-0.707106 - (-0.69635)| = 0.01076.$$

- (c) $f^{(4)}(x) = (\pi/2)^4 \sin(\pi x/2)$. So $|f^{(4)}(x)| \leq (\pi/2)^4 \approx 6.08807$ for all x . Then Thm 3.2 gives an error bound of 0.0158543. (Note: this is acceptably sharp, but with care, and since one has asked for the error at a particular point, one could get a sharper bound of 0.01121. Try it, if interested).

Q2. (Exer 3.2 from slides) Let $f(x) = \ln(x) - x^4$. Let l and S be the piecewise linear and Hermite cubic spline interpolants (respectively) to f on $N + 1$ equally spaced points $1 = x_0 < x_1 < \dots < x_N = 2$. What value of N would you have to take to ensure that

- (i) $\max_{1 \leq x \leq 2} |f(x) - l(x)| \leq 10^{-4}$?

Answer: From Thm 1.3 of Section 2.1, the error is bounded as

$$\|f - l\|_{\infty} \leq \frac{h^2}{8} \|f''\|_{\infty}.$$

Since $f''(x) = -2(6x^2 + x^{-2})$ is negative and decreasing for on $1 \leq x \leq 2$, $\|f - l\|_{\infty} = -f''(2) = 97/2 = 48.5$. So we need to choose h so that $(h^2)(48.5)/8 \leq 10^{-6}$. That gives $h \leq \sqrt{8 \times 10^{-6}/48.5} = 4.06 \times 10^{-4}$. Since $N = 1/h$, this gives $N \geq 2462.2$. As N must be an integer, we choose $N = 2463$.

(ii) $\max_{1 \leq x \leq 2} |f(x) - S(x)| \leq 10^{-4}?$

Answer: From Thm 3.2 of Chapter 3, the error is bounded as

$$\|f - S\|_{\infty} \leq \frac{h^4}{384} \|f^{(iv)}\|_{\infty}.$$

Since $f^{(iv)}(x) = -12(2 + x^{-4})$ is negative but increasing for on $1 \leq x \leq 2$, $\|f - l\|_{\infty} = -f''(1) = 36$. So we need to choose h so that $(h^4)(36)/384 \leq 10^{-6}$. That gives $h \leq (384 \times 10^{-6}/36)^{1/4} = 5.715 \times 10^{-2}$. Since $N = 1/h$, this gives $N \geq 17.498$. As N must be an integer, we choose $N = 18$.

Q3. (Based on Exer 3.3.3). There are ways of constructing the PCHIP, other than that shown in (1) of Section 3.2 above. For example, let $t = x - x_{i-1}$, then

$$S_i(x) = \frac{h^3 - 3ht^2 + 2t^3}{h^3} f_{i-1} + \frac{3ht^2 - 2t^3}{h^3} f_i + \frac{t(t-h)^2}{h^2} f'_{i-1} + \frac{t^2(t-h)}{h^2} f'_i,$$

Show that this is the same as the PCHIP.

Answer: I think the easiest approach to this is not to try to derive this formula, but just to check that it has the appropriate properties. Clearly $S_i(x)$ is a cubic polynomial in x . Next we need to verify that

$$(i) S_i(x_i) = f_{i-1}, \quad (ii) S_i(x) = f_i, \quad (iii) S'_i(x_i) = f'_{i-1}, \quad (iv) S'_i(x_i)' = f_{i-1}.$$

I just show how to do (i) and (iv). First note, that if $x = x_{i-1}$, then $t = 0$. Therefore

$$S_i(x_{i-1}) = \frac{h^3 - 3h(0^2) + 2(0)}{h^3} f_{i-1} + \frac{3h(0^2) - 2(0^3)}{h^3} f_i + \frac{0(-h)^2}{h^2} f'_{i-1} + \frac{(0^2)(-h)}{h^2} f'_i = f_{i-1},$$

which gives (i). For (iv), we first differentiate to get

$$S'_i(x) = \frac{-6ht + 6t^2}{h^3} f_{i-1} + \frac{6ht - 6t^2}{h^3} f_i + \frac{3t^2 - 4ht + h^2}{h^2} f'_{i-1} + \frac{3t^2 - 2ht}{h^2} f'_i,$$

Note that, if $x = x_i$ then $t = h$. So

$$S'_i(x_i) = \frac{-6h^2 + 6h^2}{h^3} f_{i-1} + \frac{6h^2 - 6h^2}{h^3} f_i + \frac{3h^2 - 4h^2 + h^2}{h^2} f'_{i-1} + \frac{3h^2 - 2h^2}{h^2} f'_i = f'_i,$$

as required.

Q4. Let $q_0, q_1, \dots, q_N$ be the quadrature weights for the Newton-Cotes rule $Q_N(f)$. For simplicity, let's assume that the quadrature rule is integrating f on the interval $[-1, 1]$. Show that $q_i = q_{N-i}$ for $i = 0, 1, \dots, N$.

Answer: There are a few possible ways of answering this one. Here is one. Recall that $q_i = \int_{-1}^1 L_i(x)dx$, where L_i is the i th Lagrange polynomial associated with the points $-1 = x_0 < x_1 < \dots < x_n = 1$. That is, $L_i(x)$ and $L_{n-i}(x)$ are the unique polynomials of degree n with the properties that

$$L_i(x_j) = \begin{cases} 1 & x_j = x_i \\ 0 & x_j \neq x_i, \end{cases} \quad \text{and} \quad L_{n-i}(x_j) = \begin{cases} 1 & x_j = x_{n-i} \\ 0 & x_j \neq x_{n-i}. \end{cases}$$

Since the x_i are uniformly spaced on $[-1, 1]$ we can see that $x_i = -x_{n-i}$. Therefore,

$$L_{n-i}(-x_j) = \begin{cases} 1 & x_j = -x_{n-i} = x_i \\ 0 & x_j \neq -x_{n-i} = x_i. \end{cases} \quad \text{Thus } L_{n-i}(x) = L_i(-x). \quad \text{With the substitution } y = -x, \text{ we can}$$

see that $q_{n-i} = \int_{-1}^1 L_{n-i}(x)dx = \int_{-1}^1 L_i(-x)dx = -\int_1^{-1} L_i(y)dy = \int_{-1}^1 L_i(y)dy = q_i$ (note the change in the limits of integration). So $q_i = q_{n-i}$.

Q5. Let $q_0, q_1, \dots, q_N$ be the quadrature weights for the Newton-Cotes rule $Q_N(f)$ for estimating $\int_a^b f(x)dx$.

Show that $\sum_{i=0}^N q_i = b - a$.

Answer: This comes from the fact that the, for any N , the method is exact for the constant function $f(x) \equiv 1$.